Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · LLM Evaluation · AI Observability · Guardrails · VectorDB · RAG · Python · AWS · LangGraph · Responsible AI · 9yr Production

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PROFESSIONAL SUMMARY

Senior AI engineer with 9+ years of production experience specializing in LLM evaluation, AI observability, guardrails, and responsible AI systems - the exact capabilities Capital One's IFX team is scaling across the organization. AgentProbe is the direct proof: I designed and shipped a production AI evaluation platform end-to-end - 5-metric quality engine measuring correctness, groundedness, hallucination rate, latency, and token efficiency; LLM-as-Judge scoring with pass@k multi-trial harness; structured tracing across all agent steps; CI-integrated regression gate blocking quality drift before deployment; and offline batch evaluation pipelines for large-scale analysis. This is model evaluation, experimentation, governance, and observability - the core of what IFX builds. LeaseI demonstrates responsible AI at the production level: deterministic HITL guardrails enforcing safe execution boundaries for regulated decisions, confidence-based routing, and structured output validation at every agent boundary - built with ChromaDB vector database and RAG pipelines. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime on AWS infrastructure (S3, EC2, RDS, Glue), giving me the production engineering rigor that separates AI engineers who ship from those who prototype. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026) - Virginia local, available immediately.

TECHNICAL SKILLS

AI Engineering & Evaluation: Python (9yr), LangGraph, LangChain, LLM Evaluation, Model Evaluation, AI Observability, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, Experimentation, Governance, Responsible AI, Guardrails, HITL Controls, Output Validation, Confidence Scoring, Prompt Engineering, Multi-Provider LLM Routing, Anthropic API, OpenAI API, Ollama, PyTorch (ML pipelines), AI/ML Algorithms, AI Services Deployment

Similarity Search, RAG & Data Systems: VectorDB (ChromaDB), Similarity Search, Semantic Search, Embeddings, RAG Pipelines, FastAPI, REST APIs, Async Processing, RabbitMQ, Microservices, PostgreSQL, MySQL, Oracle, Redis, High-Throughput Systems, Performance Optimization, Query Optimization, Distributed Systems, Reliability Engineering, Idempotency

Cloud, Infrastructure & Observability: AWS (S3, EC2, RDS, Glue, Lambda), Docker, GitHub Actions CI/CD, Structured Logging, Distributed Tracing, Monitoring, Alerting, Incident Response, Root-Cause Analysis, Runbooks, Automated Testing, Regression Testing, Design Documents, Technical Leadership, Code Reviews, Mentoring, Agile/Scrum, Linux, Bash, Git

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production AI Systems Delivery

- Designed and deployed production AI system (LangGraph, FastAPI, Python) as sole technical owner - built evaluation pipeline, CI validation gate, HITL guardrails for regulated decisions, and observability infrastructure; delivered to production in weeks with measurable outcomes driving iteration decisions throughout.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS cloud infrastructure; designed observability dashboards improving research comprehension 40%; cross-functional technical delivery across 3 research pipelines; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production AI-ready backend systems** - owned mission-critical Python backend systems for 7 years at 99.5% uptime on AWS infrastructure (S3, EC2, RDS, Glue); RabbitMQ event-driven async processing with retry logic and idempotency guarantees; 200+ PostgreSQL/MySQL/Oracle query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD with automated regression testing; 100% PHI compliance on AWS S3 data migration.
- **Observability, governance & responsible systems** - built structured logging, monitoring, and alerting infrastructure for production oversight; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%; maintained zero compliance violations across full 7yr ownership - the responsible AI engineering culture Capital One requires at enterprise scale.
- **Technical leadership** - wrote design documents and architecture decisions for 20+ production modules; led projects spanning multiple engineers; mentored 5+ engineers on system design and production best practices; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: LLM Evaluation & AI Observability Platform - Responsible AI at Scale

LLM Evaluation · Observability · Guardrails · Governance · CI Gate · Open Source

Python · FastAPI · LangGraph · ChromaDB · Redis · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation and observability platform - the responsible AI infrastructure Capital One's IFX team operationalizes across the enterprise. **Evaluation engine:** 5-metric quality measurement (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring calibrated against human judgments; pass@k multi-trial harness surfacing variance in non-deterministic AI outputs - the experimentation framework for responsible AI at scale. **Governance & observability:** structured tracing across all AI pipeline steps with full span capture; monitoring and alerting on quality drift; feedback loops from production back into evaluation suites; audit-ready evaluation records. **CI regression gate:** GitHub Actions integration blocks quality degradation before deployment - zero regressions across 1,240+ test cases. **Multi-provider routing:** Anthropic, OpenAI, Ollama with cost and performance tradeoffs. Open-source, multi-team deployable.

Leasel: Responsible AI System - Guardrails, VectorDB & RAG for Regulated Decisions

Responsible AI · Guardrails · HITL · VectorDB · RAG · Production

Python · LangGraph (8-agent) · FastAPI · ChromaDB · RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production responsible AI system demonstrating the guardrails and governance architecture Capital One's IFX team builds for regulated financial decisions. **Guardrails:** deterministic HITL escalation enforces human oversight on every regulated decision path - LLM reasons, deterministic logic decides; confidence-based routing avoids AI overreach on low-confidence inputs. **Similarity search & VectorDB:** ChromaDB semantic retrieval over regulatory knowledge corpus; citation-backed outputs with groundedness scoring ensure every AI response is traceable to authoritative source. **Responsible design:** structured output validation at every agent boundary; audit trail on all escalation decisions. **Measured outcomes:** 93.6% precision, 87.9% F1 across 1,240 test cases - measurement-driven responsible AI.

AI on FHIR: AI Services on AWS - Compliance, Governance & Responsible Design

AWS · AI Services · HIPAA · Compliance · Responsible AI · Production

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Production AI services on AWS for a regulated healthcare environment - LLM-powered tool invocation with HIPAA-compliant RBAC, full data isolation, and audit trail; AWS Lambda serverless backend with API gateway; PostgreSQL data services; complete observability stack. Demonstrates the responsible AI engineering practices - compliance, governance, auditability, and monitoring - that Capital One requires across all AI-powered customer experiences. ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87 · Virginia local

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017